



Interview Q&A; — Topic 2

8 Questions • Beginner → Expert

● Beginner

● Intermediate

● Expert

React Interview Prep • Topic 2: JSX • Akshay Chavhan

Interview Questions & Model Answers

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Beginner

Q1. What is JSX?

→ JSX is a syntax extension for JavaScript that looks like HTML but is compiled by Babel into `React.createElement()` calls. The browser never sees JSX — it sees plain JavaScript objects that form the Virtual DOM.

■ *Interview Tip: Always mention: JSX → Babel → React.createElement() → JS Object → Virtual DOM*

2

Beginner

Q2. Is JSX mandatory in React?

→ No. You can write `React.createElement()` directly. But JSX makes it far more readable. In React 17+, you don't even need to import React at the top of every file — the new JSX transform handles it automatically.

■ *Interview Tip: Bonus: Mention React 17+ automatic JSX transform — shows you know the evolution.*

3

Beginner

Q3. Why `className` instead of `class` in JSX?

→ Because JSX is JavaScript under the hood, and `class` is a reserved keyword in JS for defining classes. To avoid conflicts, React uses `className` for CSS classes. Similarly, `for` is replaced with `htmlFor` in labels.

■ *Interview Tip: Always pair `className` with `htmlFor` — interviewers often ask both together.*

4

Intermediate

Q4. Why can't you use if statements inside JSX?

→ JSX curly braces `{}` only accept JavaScript expressions — things that evaluate to a value. `if` is a statement, not an expression. So we use ternary operators (`condition ? a : b`) or short-circuit `&&` instead.

■ *Interview Tip: Expression = produces a value. Statement = performs an action. Know this distinction cold.*

5

Expert

Q5. What is `$$typeof` and why does it exist?

→ It's a Symbol that React attaches to every element for XSS security. Since JSON cannot serialize Symbols, a malicious server-injected object can never fake being a valid React element — React will refuse to render it.

■ *Interview Tip: This is a senior-level question. Most candidates don't know it — saying this will impress.*

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Intermediate

Q6. What does Babel do with JSX?

→ Babel is a JavaScript compiler that transforms JSX syntax into `React.createElement()` function calls that browsers can understand. Without Babel (or a similar tool like SWC), browsers would throw a syntax error on JSX.

■ *Interview Tip: SWC is the newer, faster Rust-based alternative to Babel used in Vite/Next.js — good to mention.*

7

Beginner

Q7. What is a Fragment and why use it?

→ A Fragment (`<>` or `<Fragment>`) lets you return multiple elements from a component without adding an extra DOM node. This keeps the DOM clean and avoids unnecessary wrapper divs that break CSS layouts like Flexbox/Grid.

■ *Interview Tip: Mention the short syntax `<>` vs `<Fragment>` — Fragment supports key prop, `<>` does not.*

8

Beginner

Q8. Can you use JavaScript expressions inside JSX? Give examples.

→ Yes — anything inside `{ }` is evaluated as a JS expression. Examples: `{name}` for variables, `{isLoggedIn ? : }` for conditionals, `{items.map(i => {i})}` for lists, `{getGreeting()}` for function calls.

■ *Interview Tip: The `.map()` for lists example is extremely common in interviews — always have it ready.*